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ii. Photo Chemical Phase:

- Photolysis of water and oxygen evolution
- Electron transport and synthesis of assimilatory power.

2. Dark reaction (Biosynthetic phase): Fixation and reduction of CO_2 into carbohydrates with the help of assimilatory power produced during light reaction. This reaction does not require light and is not directly light driven. Hence, it is called as **Dark reaction or Calvin-Benson cycle** (Figure 13.10).

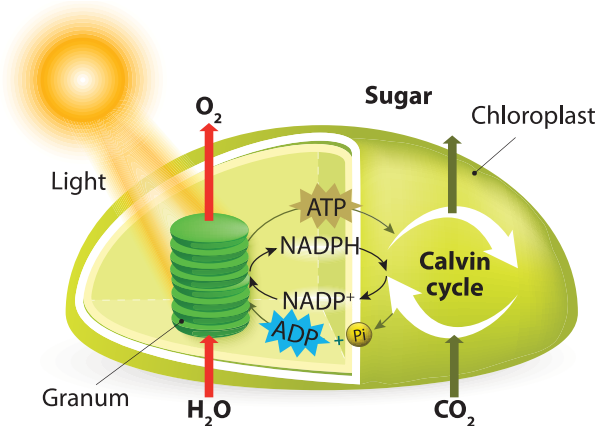


Figure 13.10: Light and Dark Reaction

13.9 Photo-Oxidation Phase of Light Reaction

The action of photon plays a vital role in excitation of pigment molecules to release an electron. When the molecules absorb a photon, it is in excited state. When the light source turned off, the high energy electrons return to their normal low energy orbitals as the excited molecule goes back to its original stable condition known as **ground state**. When molecules absorb or emit light they change their electronic state. Absorption of blue light excites the chlorophyll to higher energy state than absorption of Red light, because the energy of photon is higher when their wavelength is shorter. When the pigment molecule is in an excited state, this excitation energy is utilised for the phosphorylation. Phosphorylation takes place with the help of light generated electron and hence it is known as **photophosphorylation**.

13.9.1 Fluorescence and Phosphorescence

Normal state of an atom or molecule is called **ground state**. When a photon of light collides with the chlorophyll molecule, an electron from outer most orbit is moved to higher energy orbit causing excitation of chlorophyll. This is known as **excited state**. There are three excited states such as:

1. First singlet state (S_1)
2. Second singlet state (S_2)
3. First Triplet Sate (T_1)

When a red light strikes chlorophyll molecule, one electron is released from its ground level (S_0) to first singlet state (S_1). It is in unstable state having half-life period of 10^{-9} seconds. When a blue light strikes chlorophyll molecule, one electron is released from its ground level (S_0) to second singlet state (S_2). It is because blue light has shorter wavelength and more energy than red light. This state is also unstable having half-life period of less than 10^{-12} seconds. Both S_1 and S_2 states being unstable move to ground state S_0 by releasing energy through the several possible ways.

i. Fluorescence

The electron from first singlet state (S_1) returns to ground state (S_0) by releasing energy in the form of radiation energy (light) in the red region and this is known as **fluorescence**. Fluorescence is the immediate emission of absorbed